



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment-4

Experiment No.: 4

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code, Google Chrome

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker</title>
  <style>
    body {
      font-family: Georgia, sans-serif;
      background: #f8f2d9;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
    }
    .container {
      background: white;
      padding: 25px;
      border-radius: 10px;
      width: 350px;
      box-shadow: 0px 0px 10px rgb(27, 21, 21);
      text-align: center;
    }
    input {
      width: 90%;
      padding: 10px;
      margin: 10px 0;
      font-size: 16px;
    }
    button {
      padding: 10px 20px;
      background: royalblue;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <div class="container">
    <input type="text"/>
    <button>Check</button>
  </div>
</body>
</html>
```



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```
button:hover {
  background: darkblue;
}
#result {
  margin-top: 20px;
  font-size: 18px;
  font-weight: bold;
}
#counter {
  color: green;
  margin-top: 10px;
}
</style>
</head>
<body>
<div class="container">
  <h2>Palindrome Checker</h2>
  <input type="text" id="word" placeholder="Enter a word">
  <br>
  <button onclick="checkPalindrome()">Check</button>
  <p id="result"></p>
  <p id="counter"></p>
</div>
<script src="script.js"></script>
</body>
</html>
```

Script.js:

```
let appName = "Palindrome Checker";
function visitCounter() {
  let count = 0;
  return function () {
    count++;
    return count;
  };
}
const counter = visitCounter();
function isPalindrome(text) {
  let cleaned = text.toLowerCase().replace(/[^a-z0-9]/g, "");
  let reversed = cleaned.split("").reverse().join("");
  return cleaned === reversed;
}
const displayMessage = function(message, color) {
  const result = document.getElementById("result");
```



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```
result.innerHTML = message;
result.style.color = color;

};
const updateCounter = () => {
    document.getElementById("counter").innerHTML =
        "Checked " + counter() + " time(s)";
};
function checkPalindrome() {
    try {
        let input = document.getElementById("word").value;
        if (input.trim() === "") {
            throw "Input cannot be empty.";
        }
        if (isPalindrome(input)) {
            displayMessage(
                `"{input}" is a Palindrome`,
                "green"
            );
        } else {
            displayMessage(
                `"{input}" is NOT a Palindrome`,
                "red"
            );
        }
        updateCounter();
    }
    catch(error) {
        displayMessage(error, "orange");
    }
}
```



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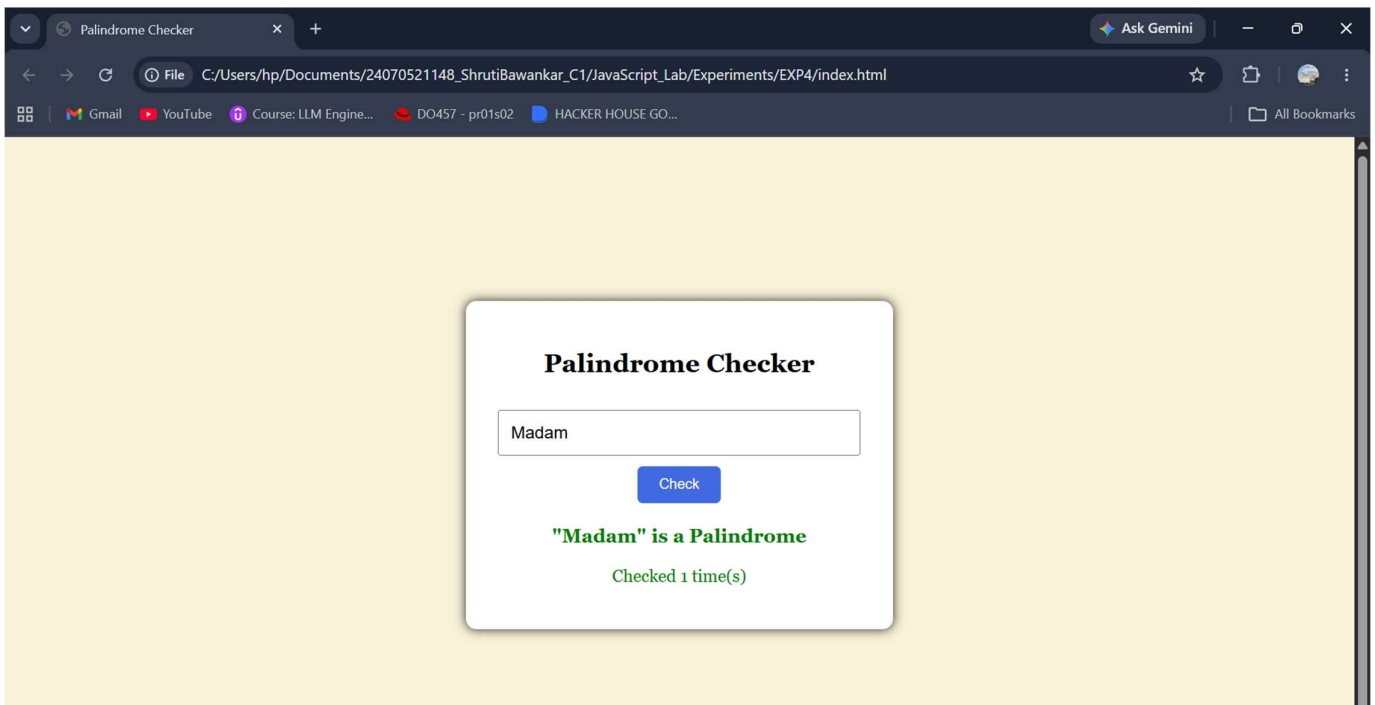
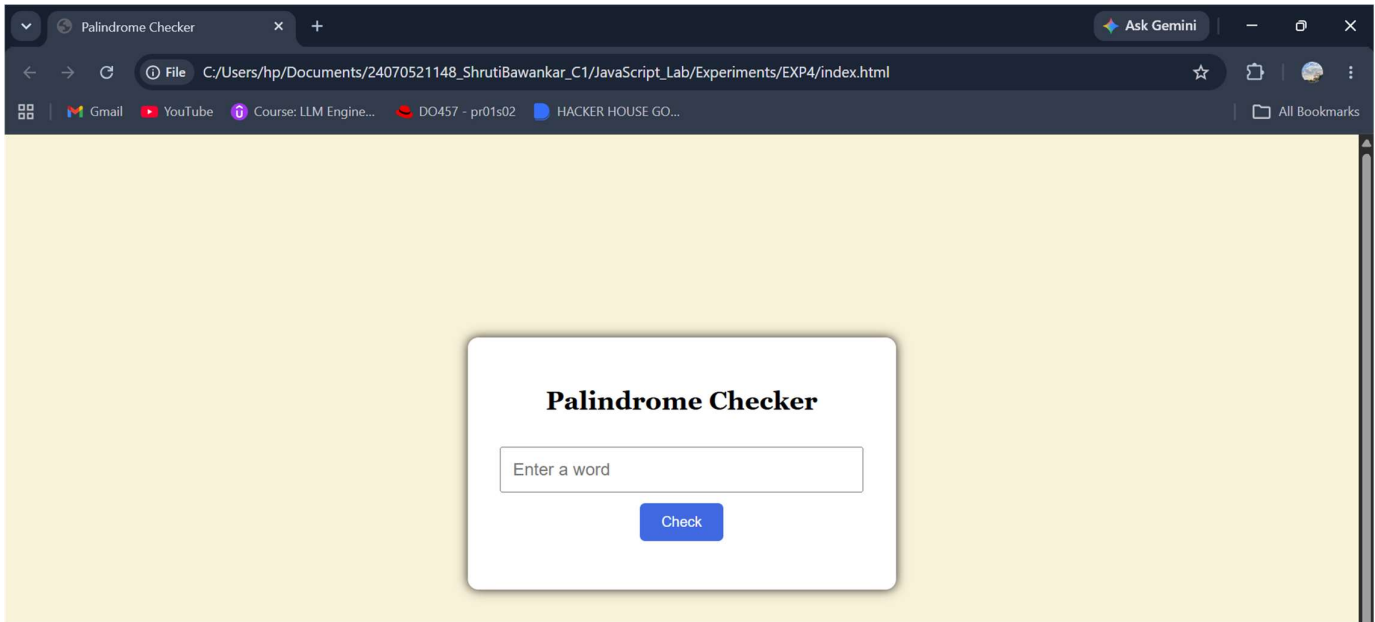
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Output:





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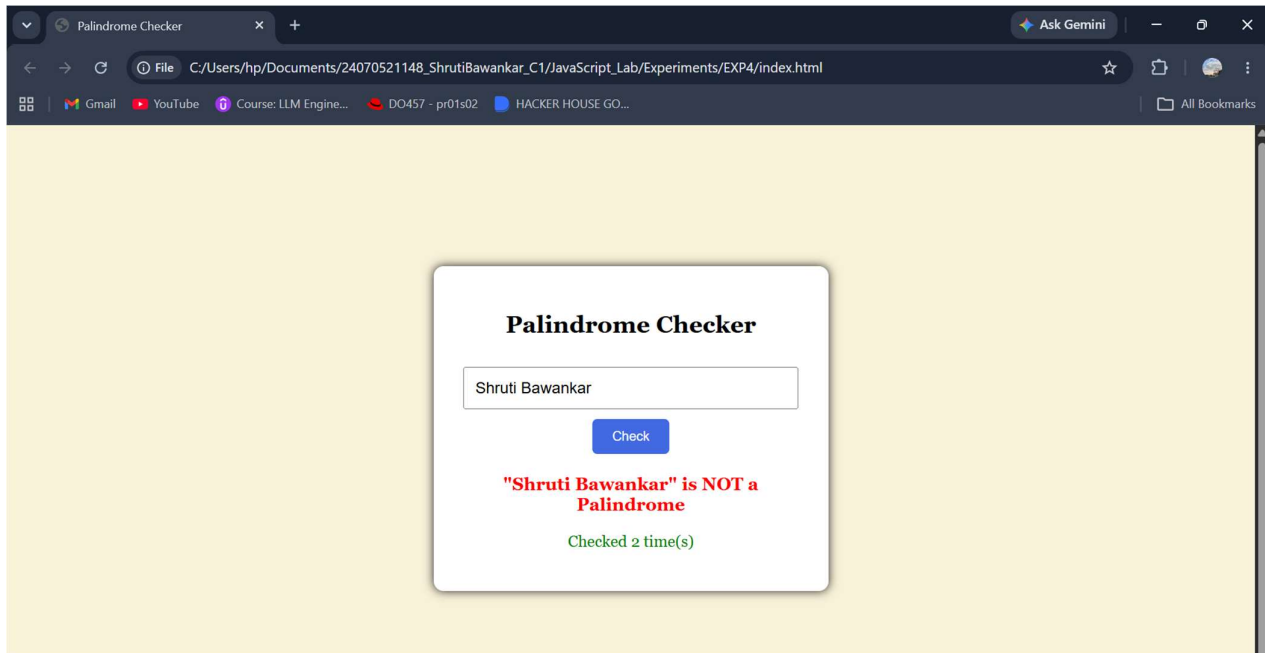
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Result: The palindrome checker successfully identifies whether the entered text is a palindrome or not and displays the result while handling invalid input using try...catch.

Conclusion: The program effectively demonstrates JavaScript function types, scope, closures, and exception handling through a simple and user-friendly palindrome checking application.